

TOM ALMOG

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EDUCATION

University of Waterloo

Sep 2025 - May 2030

- *Bachelor of Computer Science Co-op*

EXPERIENCE

Machine Learning Engineer - Parkalytics

May 2026 - Present

- Automating vehicle counts from orthomosaic drone imagery by developing computer vision models, reducing manual review.
- Building a parking-stall detection model that predicts valid spaces with **90%+** accuracy, reducing processing time by over **60%**.
- Engineering cloud pipelines with **AWS S3, Lambda, and Aurora** to store, process, and serve interactive aerial imagery to clients.

Data Engineer Intern - Polymarket

Apr 2026

- Built an **autoencoder** to flag potential insider trading based on anomalous signals, greatly reducing human investigation time.
- Engineered features from historical trades & performance, ranking suspicious activity by reconstruction error for analyst review.
- Reduced false-positive alerts by **20%** through market-specific anomaly thresholds, preventing legitimate users from being flagged.

Machine Learning Engineer - WatStreet

Dec 2025 - Mar 2026

- Built a **Transformer-based** time series model to improve market-making backtest performance by **25%** over a static baseline.
- Reduced model experimentation time by building a reusable pipeline for sequence generation & training using **Pandas & PyTorch**.
- Integrated model predictions into a regime-aware quoting engine that adapted trading behavior as market conditions changed.

Machine Learning Engineer - Wat.AI [Research Paper]

Sep 2025 - Dec 2025

- Designed **HydraLA-Net**, a segmentation model with **4 prediction heads** to improve retinal lesion segmentation accuracy.
- Built a training pipeline using **3 datasets**, ablating CLAHE preprocessing and class-imbalance-aware Focal Tversky/BCE losses.
- Improved microaneurysm recall by **15.4%** & mean F1 by **3.2%** through architectural, preprocessing, & loss-function optimization.

Autonomous Systems Engineer - Waterloo Aerial Robotics Group

May 2025 - Aug 2025

- Trained **deep-learning** models for object detection, tracking, and scene segmentation in autonomous UAV perception.
- Built preprocessing pipelines for vision datasets of **200K+** images to improve model generalization and robustness during flight.
- Optimized and deployed perception models to onboard UAV systems, achieving real-time inference at over **45 FPS**.

PROJECTS

Crucible, Agentic Model Improvement Platform (Python + PyTorch + Tauri + React) [GitHub] [Live]

Feb 2026 - Present

- Built an **LLM fine-tuning & evaluation** platform supporting 13 training methods, 4,000+ benchmarks, and custom evals.
- Developed an **AI agent with 35+ MCP tools** to diagnose model failures, launch training jobs, and compare results.
- Orchestrated **PyTorch** workloads across local, SSH, and Slurm GPUs with experiment tracking and reproducibility.

Retinal OCT Disease Classifier (PyTorch + EfficientNet + Grad-CAM) [GitHub] [Hugging Face]

Jan 2026

- Fine-tuned **EfficientNet-B3** on 84,000+ retinal OCT images, achieving **99.6% test accuracy** across 4 classes.
- Built an FP16 **PyTorch** pipeline with data augmentation and Grad-CAM, then published the model to Hugging Face.

CarbonAware-ML (Python + PyTorch + Electricity Maps API) [GitHub] [PyPI]

Sep 2025

- Built a **PyTorch scheduler** that pauses and resumes training using live carbon-intensity and electricity-price signals.
- Integrated Electricity Maps and WattTime across 230+ regions with CLI/TensorBoard support, reaching **2,400+ PyPI downloads**.

PUBLICATION

T. Almog, *An Analysis of Optimizer Choice on Energy Efficiency and Performance in Neural Network Training*

[GitHub] | [SSRN:5465242] | [arXiv:2509.13516]

TECHNICAL SKILLS

Languages: Python, C, C++, C#, Java, Rust, Go, JavaScript, TypeScript, SQL, Bash, HTML/CSS

ML & Data: PyTorch, TensorFlow, Keras, scikit-learn, Hugging Face Transformers, OpenCV, torchvision, NumPy, Pandas, SciPy

Technologies: AWS (S3, Lambda, Aurora), Docker, Linux, SSH, Slurm, MCP, FastAPI, PostgreSQL, SQLite, Supabase, React, Next.js, Node.js, Tauri, Git